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Summary of Structured File, Streaming Input / Output

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Structured Files

A single C file can read, write, move, and create files in our computer easily using a few functions and elements included in the C File I/O system. We can easily manipulate data in a file regardless of whether the file is a text file or a binary file using functions like fopen(), fclose(), fprintf(), fscanf(), getc(), putc(), getw(), fseek(), etc.

What are files in C?

In C programming, files allow us to save information so that it can be accessed and used later. C provides different functions that allow programmers to create, open, read, and write, and close files. These functions make it possible to store and manage information outside of the program itself.

Types of Files in C

There are mainly two types of files that can be used in C programming:

- Text Files store information in a readable format using characters. They usually use the .txt file extension. They can be opened, viewed, and edited using a regular text editor.
- Binary Files store information in thr form of bytes rather than readable text. They represent information using binary data (0s and 1s). They are commonly used to store numerical information and other types of data.

Streaming input & output

In C programming, a stream is a way for data to move between a program and another device, such as a keyboard, computer screen, or file. It alloes programmers to handle input and output without having to worry about how the hardware works.

There are three types of main streams:

Input Stream – Allows a program to receive information from an outside source.

, **Output Stream** – Allows a program to send information to another location, such as the screen or a file.

Error Stream – Used to display error messages when something goes wrong during program execution.

Three Standard Streams in C

C automatically provides three standard streams whenever a program starts running:

Standard Input (stdin): Receives information from the user, usually through the keyboard.

Standard Output (stdout): Displays information from the program, usually on the computer screen.

Standard Error (stderr): Displays error messages to help identify problems within a program.

Necessary Libraries

C programming uses libraries to provide functions that help programs perform different tasks. These libraries make it easier for programmers to work with files, receive input, display information, and manage data. Without these built-in functions, programmers would have to create many of these operations themselves. C itself cannot print, read a file, or allocate memory, so all of that comes from the standard library. You could also add that the library is defined by the ISO/IEC 9899 standard.

stdio.h: This library provides the main input and output functions used in C. It includes functions such as printf(), scanf(), fopen(), and fclose() that allows programs to read and write information.

stdlib.h: This library provides general functions that help with memory management, number conversions, searching, and sorting information.

string.h: This library contains functions used to work with strongs. It allows programmers to compare, combine, and modify text.

It handles string manipulation, such as concatenation and comparison. It's handy for preparing text before writing it to a file.

errno.h: This library helps identify errors that happen when certain functions fail. It provides error information that can help programmers understand what went wrong.

File pointers

- What is a File Pointer?
- A file pointer is a special variable in C that allows a program to access and manage an opened file. It helps the program kee p track of where it is currently working within a file, whether it is reading or writing information.
- How Does It Work?
- The fopen() function is used to open a file and create a file pointer.
- The file pointer keeps track of the current position within the file.
- Every time information is read or written, the file pointer's position is updated.
- The fclose() function closes the file when the program is finished using it.

Syntax

*FILE *fp;* FILE is a predefined structure declared in <stdio.h>. fp is a pointer that stores the address of an opened file.

Purpose and Limitations

File pointers help C programs keep track of important file information, such as the current position, access mode, and file status.

However, there are a few things programmers need to be careful about:

A file pointer cannot be used after its file has been closed.

If fopen() fails, it returns NULL, so the program should always check whether the file opened successfully.

Using an invalid or uninitialized file pointer can cause errors or crashes.

The internal structure of a FILE pointer should not be changed directly.

fread and fwrite

What are fread() and fwrite()?

In C programming, fread() and fwrite() are functions used to read and write binary data to files. Instead of converting information into readable text, these functions work directly with data stored in memory. They are especially useful when working with structures because they allow entire blocks of information to be saved and retrieved.

Parameters, Both functions use four main parameters:

ptr: Points to the location in memory where the data will be read from or written to.

size: Specifies the size of each piece of data in bytes.

nmemb: Determines how many elements will be read or written.

stream: Identifies the file where the operation will take place using a file pointer.

Important Things to Remember

- Use "wb" when opening a file to write binary information.
- Use "rb" when opening a file to read binary information.
- Always check whether the file opened successfully before performing operations.
- fwrite() saves binary information to a file.
- fread() retrieves binary information from a file.
- Use fclose() when finished working with the file.

fprintf

What is fprintf()?

- The fprintf() function is used in C programming to write formatted information directly into a file. It works similarly to printf(), but instead of displaying information on the computer screen, it sends that information to a file.
- It also allows programmers to use format specifiers to write different types of information, such as numbers, characters, and strings.

Syntax

*int fprintf(FILE *stream, const char *format, ...);*

Parameters

stream: Identifies the file where the information will be written.

format: Contains the text and format specifiers that determine how the information will appear.

...: Represents the additional values that will be written into the file.

Return Value

Returns the number of characters successfully written to the file. Returns a negative value if an error occurs during the operation.

Advantages of fprintf()

The fprintf() function is useful because it allows programmers to save organized information directly into files.

Allows formatted information to be written into files. Supports format specifiers such as %d, %f, %c, and %s. Can be used to create reports, save program results, and store log information.

Standard streams

What are Streams?

In programming, a stream refers to the movement of data between a program and another source or destination. It acts as a connection that allows information to enter or leave a program. Streams make it possible for programs to communicate with devices such as keyboards, computer screens, and files.

Three Standard Streams in C

Whenever a C program starts running, three standard streams are automatically available. These streams are included through the stdio.h library.

- Standard Input (stdin): Allows a program to receive information from an outside source, usually through the keyboard.
- Standard Output (stdout): Sends information from the program to a destination, usually the computer screen.
- Standard Error (stderr): Used to display error messages when problems occur during program execution.

How Are They Used?

These three streams allow programmers to control how information enters and leaves a program. They can also be redirected to work with different files or processes instead of using the default keyboard and screen.

Input / Output Redirection

What is Input / Output Redirection?

In C programming, input and output redirection allows programmers to change where a program gets its information and where it s results are sent. Instead of always using the keyboard for input and the computer screen for output, a program can use files or other locations.

Standard Input Redirection

Standard input can be changed so that a program reads information from a file instead of receiving it directly from the keyboard. This is useful when working with saved data or testing a program.

// Redirect input from a file ./myprogram < input.txt

The < symbol tells the program to use input.txt as its input source.

Standard Output Redirection

Standard output can also be redirected so that information is saved to a file instead of appearing on the computer screen. This is useful for saving program results, creating reports, and keeping records.

// Redirect output to a file ./myprogram > output.txt

The > symbol tells the program to send its output to output.txt.

SOURCES

C - File I/O - GeeksforGeeks

What is Stream in C Programming? - Sanfoundry

Basic Input and Output in C - GeeksforGeeks

C programming/Stream IO - Wikibooks, open books for an open world

C File Pointer - GeeksforGeeks

Read/Write Structure From/to a File in C - GeeksforGeeks

Explain the functions fread() and fwrite() used in files in C

fprintf() in C - GeeksforGeeks

Standard Streams (The GNU C Library)

Streams in Computer Programming | Definition, Types & Examples - Lesson | Study.com

Conclusion

Structured files and input/output streams are important parts of C programming because they allow programs to store, access, and manage information. Text files save data in a format that people can easily read, while binary files store information in a format that computers can process directly. File pointers help programs keep track of open files and manage different file operations.

Functions such as fprintf(), fread(), and fwrite() make it possible to save and retrieve information in different formats. Standard streams like stdin, stdout, and stderr allow programs to receive input, display output, and report errors. Input and output redirection also gives programmers the ability to read information from files or save program results to them.

Overall, these tools make C programs more flexible and efficient when working with information. Understanding how they work helps programmers create programs that can store, organize, and process data more effectively.